CSE3211: Software Project Planning and Management
Class Test #1(29.04.2024)

1. Address some tips for leading a successful kick-off meeting.
2. What is a work breakdown structure? How to create a work breakdown structure. Write some tips for making a work breakdown structure.
3. In this activity, you will identify a project's major milestones, break them into smaller tasks, and complete a Work Breakdown Structure (WBS) brainstorm diagram.
   Setting tasks and milestones gives you a clear understanding of the amount of work your project will require, so you can keep your project on track. Milestones also serve as great check-in points to highlight the project's progress for your team and stakeholders.

Scenario

As a project manager for Office Green, your job includes working with the operations team to develop and implement an Operations and Training plan. Your team will use this plan for ongoing maintenance of the Plant Pals service. It will take six months to fully implement all protocols, including setting up operational tools, putting delivery processes in place, and training employees. The project will begin before the Plant Pals launch and extend beyond it.

The Operations team is responsible for managing the organization's day-to-day business so that it continues to run smoothly. Operations provide resources to other departments, ensure consistency, and maintain the company's profitability. The Chief Operations Officer (COO) typically leads the Operations team and works closely with other divisions, such as Sales, Client Services, Human Resources, Information Technology (IT), as well as the project manager.

Developing the Operations and Training plan marks a new stage of the project, with its own set of goals and deliverables. After assessing the requirements for the plan, you determine three major milestones:

- Establish a plant delivery and logistics plan.
- Select and install supply chain management software and equipment.
- Develop and launch an employee training program.

For each milestone, you divide the work into three separate tasks. Each task has its owner, duration, and details. The tasks associated with your milestones include:

- **Establish a plant delivery and logistics plan**: Team members will need to source materials for packaging and hire delivery drivers. They are also responsible for calculating the delivery fees.
- **Select and install supply chain software and equipment**: Team members will supervise vendor setup of inventory management and fulfillment software. They must also supervise vendor installation of fulfillment equipment and determine internal safety protocols for the equipment.
- **Develop and launch an employee training program**: Team members will need to develop training sessions, train employees to use the software and equipment, and monitor progress and improve training processes.

Step-by-step Instructions

Step 1: Title your chart

Title your chart "Operations and Training Plan." Giving your chart a descriptive title helps you focus on the project's milestones and tasks.

Step 2: Add milestones.

Record the three milestones from the scenario in the boxes labeled **Milestone 1**, **Milestone 2**, and **Milestone 3**.

Step 3: Add tasks.

Identify the tasks required to reach each milestone. Record three tasks for each milestone in the **Task 1**, **Task 2**, and **Task 3** boxes. When you finish, you should have nine tasks in your brainstorm diagram—three for each milestone.

Be sure to address the following points in your completed WBS brainstorm diagram:

- A descriptive title
- Three different milestones
- Nine tasks in total (three for each milestone)

4. In this activity, you will use a spreadsheet-like table to assign task owners and sort milestones and tasks into sequential order.

Scenario

As a project manager for Office Green, you are developing an Operations and Training plan for the new Plant Pals service. It will take six months to fully implement all protocols, including setting up operational tools, putting delivery processes in place, and training employees. Here are the milestones and their associated tasks:

- **Establish a plant delivery and logistics plan**: Team members will need to source materials for packaging and hire delivery drivers. They are also responsible for calculating the delivery fees.

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- **Select and install supply chain software and equipment**: Team members will supervise vendor setup of inventory management and fulfillment software. They must also supervise vendor installation of fulfillment equipment and determine internal safety protocols for the equipment.
- **Develop and launch an employee training program**: Team members will need to develop training sessions, train employees to use the software and equipment, and monitor employee progress, and improve training processes.

Now you will add those tasks and milestones to a WBS spreadsheet. You will also use the sheet to record task owners and any notes about the tasks and their owners. The project sponsor for the Operations and Training plan, the Director of Operations, has put together a team for the launch of the operation. You can select task owners from among the following teams:

- A **Financial Analyst,** who tracks and calculates costs and revenue.
- A **Fulfillment Director**, who sources materials, ensures equipment is functional, and confirms fulfillment processes are correctly implemented.
- An **Inventory Manager**, who maintains inventory and oversees software installation.
- A **Human Resources Specialist**, who manages hiring and develops training protocols (with the help of the Training Manager)
- A **Training Manager**, who reports to the HR Specialist, runs the training program on the established protocols and refines training processes.
- A **Quality Assurance Tester**, who ensures product quality and determines safety protocols and best practices.

Step-by-step Instructions

Step 1: Add milestones.

First, open your WBS brainstorm diagram. Then record the three milestones from your brainstorm diagram under the **Milestone** column of your WBS Spreadsheet.

Step 2: Add tasks.

Enter the tasks from your brainstorm diagram under the **Task** column of your spreadsheet. Write each task next to the corresponding milestone and task ID number. Assigning each task to an ID number helps you organize, analyze, and search for tasks as the list grows. Task IDs are especially useful for large or complex projects.

Task ID numbers tell you two things: 1) which milestone each task relates to and 2) the sequence of the tasks. So, task 1.1 is the ID for the first task of the first milestone. Task 1.2 is the second task of the first milestone, and so on.

When you finish, you should have nine tasks in your spreadsheet—three for each milestone.

Step 3: Add task owners and notes.

To the right of the **Task** column are columns for **Owner** and **Notes**. Here, you can record important details that didn't fit into your WBS diagram.

First, review the scenario and determine which of your team members should do each task. Then enter that person's role next to the task in the **Owner** column.

Record any helpful details from the scenario in the **Notes** column. For example, including the description of the task owner's role would remind you of each person's responsibilities.

If you'd like, you can also add your best estimate of how long each task might take in the **Duration** column (this is optional).

Be sure to address the following points in your completed WBS spreadsheet:

- The three milestones from your WBS diagram
- The nine tasks from your WBS diagram
- An owner for each task
- Notes about the owner of each task, and any additional information you'd like to include.

5. In this activity, you will create a project budget to estimate the total cost of developing and implementing an Operations and Training Plan.

Scenario

As project manager for Office Green, your job includes working with the operations team to develop and implement an Operations and Training plan for the Plant Pals service. In addition to identifying the major milestones and associated tasks for this plan, you also need to estimate the costs and create a budget.

You will use this budget as a baseline throughout the project, but you should monitor spending throughout the project and adjust as needed.

The estimated costs of the milestones and their subtasks are as follows:

Milestone 1: Establish a plant delivery and logistics plan.

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- *Task 1:* Purchase delivery trucks. Purchase two delivery trucks at a cost of $15,000 per truck.
- *Task 2:* Source packaging materials. Purchase 1,500 boxes at a cost of $2 per box.
- *Task 3:* Pay delivery drivers. Pay two delivery drivers for ten days (assume eight-hour work days) at a rate of $15 per hour.

Milestone 2: Select and install supply chain management software and equipment.

- *Task 1:* Source vendor (includes setup, installation, and deployment of software and equipment systems): Fixed cost of $15,000.

Milestone 3: Develop and launch an employee training program.

- *Task 1:* Develop training sessions. Pay the HR specialist $50 per hour for ten days (assume eight-hour workdays).
- *Task 2:* Train employees to use the software and equipment. Pay the Training Manager $25 per hour for ten days (assume eight-hour workdays).
- *Task 3:* Monitor employee progress and improve training processes. Pay the Training Manager $25 per hour for another ten days (assume eight-hour workdays).

Your estimated budget to reach these milestones is $62,000. This amount includes a reserve buffer of $3,600 to account for unexpected costs. You should generally allow for a buffer of at least 5%.

Step-By-Step Instructions

Step 1: Add milestones and tasks.

Fill in the project milestones and tasks under **Milestones & Tasks**. Be sure to group each task with its associated milestone. When you're finished, your budget should include three milestones and seven tasks. We've filled in the first milestone and task for you.

Step 2: List project employees

For each task associated with a particular employee, enter the employee who should be paid for that task under **Employee Details**. Some tasks may not have any associated employees, while others could have two or more (e.g., two delivery drivers).

Note: Your spreadsheet should include the pay rates for hourly employees, like the delivery drivers. You should only list salaried employees when they work extra hours on special projects included in the scenario (e.g., the HR Specialist developing the training sessions).

Step 3: List employee hours and rates

Next, calculate the number of hours each employee will spend on each task. Enter this number under **Hours**. Now list each employee's hourly pay rate under **Rate**. Note that the spreadsheet calculates the total cost for you.

Step 4: Enter material quantities and costs.

Now review the scenario for any materials you need to purchase (e.g., trucks, boxes):

- For materials priced per unit, enter the amount needed under **Units**. Then record the cost per unit under **$/Unit**.
- For fixed-cost materials, record the price under **Fixed Cost**.

Be sure to address the following points in your completed budget:

- Three milestones and seven tasks
- Employees responsible for tasks
- Employee hours and pay rates
- Materials and costs

6. In this activity, you will apply what you have learned about procurement value to complete a Statement of Work (SoW). A SoW is a document that lays out the products and services a vendor or contractor will provide for the organization. It also describes what the contractor needs to perform the agreed-upon services.

Scenario

As part of the Plant Pals Operations and Training plan, you selected a vendor to install supply chain management software and equipment. To ensure the vendor can successfully set up the proper systems, you need to prepare a Statement of Work (SoW).

For this engagement, the SoW should contain the following elements:

- The vendor will configure inventory and fulfillment tracking software for the Office Green team. This includes auditing existing Office Green software and developing optimizations.
- The vendor will also install the new software on *all* Office Green devices and equipment (laptops, mobile devices, and hard drives).
- The vendor will install fulfillment equipment in Office Green's warehouses.

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- The vendor will create training manuals and a maintenance guide for the software and equipment.
- The vendor will meet with the Training Manager and the HR Specialist to explain the training manuals and answer questions.
- The vendor should complete all tasks within two weeks (10 business days).
- The vendor is not responsible for training other employees or ongoing maintenance.
- Office Green will pay the vendor once they have completed all of the work.

Step-By-Step Instructions

Step 1: Title the document

First, write the project name in the page header and under the Office Green logo.

Step 2: Add important stakeholders

Record important stakeholders for the project under the project name. In this case, that includes the project lead (that's you, the project manager) and the project sponsor (the Director of Product).

*Note: You won't fill in the **Revision History** until you've drafted the SoW and shared it with stakeholders. This is where you will log changes to the SoW, so don't forget to include it.*

Step 3: State the purpose

In the **Purpose** section, write a sentence or two explaining the desired outcomes for the SoW. Try to be as specific as possible about your goals for working with the vendor and how they relate to the larger aims of the Plant Pals project.

Step 4: Define the scope

Next, define the project scope. Enter the major activities the vendor will complete into the **Scope** section.

Step 5: Define out-of-scope activities

Record activities that are beyond the contract's scope in the **Out-of-scope activities** section. Defining what's out-of-scope sets expectations and minimizes potential confusion.

Step 6: Define deliverables

Describe the tangible and intangible outcomes of the vendor's work in the **Deliverables** section. Make sure to be specific and consider the overall goals of the Plant Pals project.

Step 7: Add milestones

Consider the scenario carefully and break down the vendor's work into at least three milestones. The milestones should define the major benchmarks the vendor must reach to complete the work. They are essential to tracking progress and should align with the in-scope activities. Write them in the **Milestones** section of the SoW.

Step 8: Estimate time to completion

Use the scenario above to estimate the approximate number of hours the work should take. Assume eight-hour work days and five-day work weeks.

For example, an assignment that lasts for 20 business days should take 160 hours to complete. Enter the number in the **Estimated hours for completion** section.

Step 9: Identify a due date

Select a date by which the vendor must complete all the work. Calculate this date using the start date listed in the SoW header (April 12).

Step 10: Set payment terms

All parties need to agree on when the vendor will be paid for their work. Record the terms from the scenario in the **Payment Terms** section.

Be sure to include the following elements in your completed Statement of Work (SOW):

- A purpose statement for the SoW
- A list of in-scope items
- A list of out-of-scope items
- Expected deliverables
- Major milestones
- The estimated hours needed for completion
- An estimated date of completion

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